

Case Report of a Rare Omental Gastrointestinal Stromal tumor (GIST) in Abuja, Nigeria

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ABSTRACT

The gastrointestinal stromal tumor (GIST) is a mesenchymal spindle cell or epithelioid tumor that arises primarily in the stomach and small intestine; they are found rarely in other sites especially omentum and the mesentery. Their phenotype is characterized by the expression of c-KIT (CD117), and they are derived from the interstitial cell of Cajal. This is a rare case of a 43-year-old married woman with GIST who presented with abdominal swelling among other features to the surgical unit of university of Abuja Teaching Hospital Gwagwalada. Intraoperative finding showed a mass that was located in the greater omentum. The mass was excised and sent for histopathology analysis that confirmed the diagnosis of GIST that is CD117 positive.

Keywords: GIST, Greater omentum, CD117, case report

INTRODUCTION

The presence of Extra-Gastrointestinal Stromal Tumors (EGISTs), is exceptionally uncommon, with those EGISTs originating from the omentum being particularly rare as reported at <1 % incidence.¹ The gastrointestinal stromal tumor (GIST) is a mesenchymal spindle cell or epithelioid tumor that arises primarily in the stomach and small intestine; they are found rarely in the colon, rectum, esophagus, omentum and mesentery. Their phenotype is characterized by the expression of KIT (CD117), a transmembrane tyrosine kinase receptor for stem cell factor¹ and derived from the interstitial cell of Cajal that controls gastrointestinal peristalsis and exhibit the CD117+ immunophenotypes, which is the hallmark of GIST². Primary GIST in extra gastrointestinal tissues (EGISTs) share similar histological features and immunophenotypes to classical GISTs and belong either to GISTs that have separated from the GI tract wall, or to independent growths of omentum and

mesentery cells.³ Five cases of lesser omental GISTs have been documented⁴⁻⁷ while several reports of greater omentum GIST have been reported.⁸⁻¹⁰ The median age at diagnosis is about 60 years and seen more common in men than in women, however, a total of 3% of GIST cases were diagnosed in patients under 21 years of age.¹¹⁻¹³

CASE PRESENTATION

Our patient is a 43-year-old married woman, who presented with a history of progressive abdominal distension and weight loss of 2 months duration. There was no associated vomiting but had altered bowel habit with constipation. She had not passed blood in stool and no jaundice. Examination revealed a woman who was emaciated, pale, anicteric with no peripheral lymphadenopathy. Her pulse rate was 101/min, blood pressure 110/80 mmHg, and respiratory rate of 24/min.

Abdominal examination revealed gross distension and soft to touch with no area of marked tenderness. There was a firm non mobile mass extending from the epigastrium to the left hypochondriac region that was not tender. Examination of other systems were not contributory.

Investigations done revealed packed cell volume of 23% while other full blood count parameters were essentially within normal

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reference range and peripheral blood film showed hypochromasia and microcytosis. Serum electrolytes, urea and creatinine were within normal reference range. The random blood sugar and liver function test were within normal reference range. Abdomino - pelvic ultrasound scan showed a huge complex anterior abdominal wall mass measuring 14.3 x10 cm showing areas of cystic degeneration and lobulated outline with few calcifications. The mass was seen in the anterior peritoneal space contiguous with the kidney, spleen and related structures and Doppler interrogation shows minimal flow. Liver enlarged 20.5cm, showed at least 2 hyperechoic masses measuring 5 x 3 cm and 4.3 x 4.5 cm. Intrahepatic ducts were not dilated, spleen, pancreas and bladder were normal.

She had 2 units of blood transfused following low packed cell volume and was booked for exploratory laparotomy. Intra - operative findings revealed a complex cystic mass from the greater omentum involving the posterior abdominal wall forming part of the cyst wall that has multiple nodules/distended veins. The cyst contained brownish fluid with

necrotic debris. The small bowel was displaced downwards (See figure 1). Much of cyst wall was excised and saline lavage was done only in the cyst cavity that had necrotic tissue before it was excised and drain was inserted. She had uneventful postoperative state and was placed on analgesics, antibiotics, and hematinic. The drain was active for a week before it stopped draining and the tissue mass was taken for histopathological analysis. Macroscopy showed dark grey soft to firm multinodular mass with cystic areas, measuring 13x10x7cm with cut section showing firm variegated appearance (see figure 2), while microscopy revealed neoplastic cells that were made up of fascicles of spindle to ovoid cells with ovoid nuclei and have mild atypia. Few mitosis were noted (See figure 3). The diagnosis of GIST was made histologically and confirmed by CD117 immunohistochemistry positivity (see figure 4). Patient was placed on imatinib 400mg daily and she commenced regular follow up at the surgical outpatient department following. The patient is doing well and for 2 years now, she has been on follow up with imatinib.

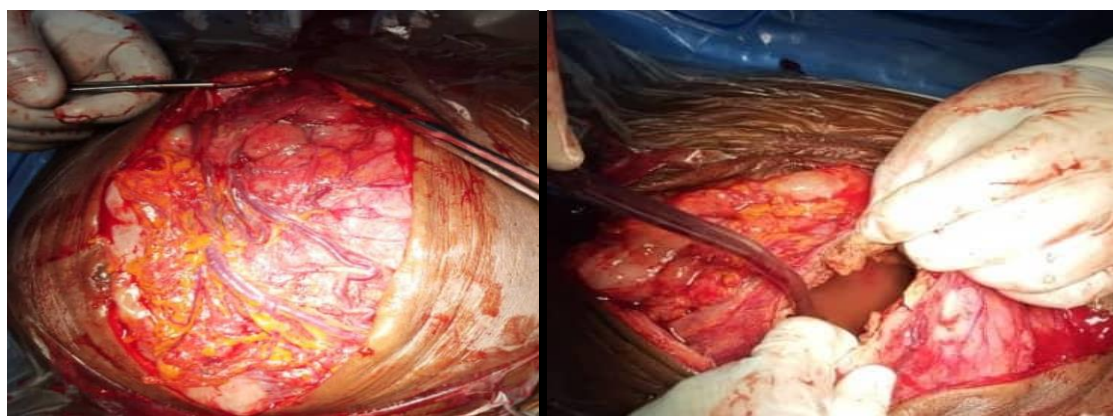


Figure 1: Intra operative finding of omental GIST with cystic areas.



Figure 2: Gross appearance showing multinodular grayish white firm mass with cystic areas and a firm cut surface.

DISCUSSION

The interstitial cells of cajal are believed to be responsible for gastrointestinal stromal tumors, which control peristalsis in the GI tract and are the only cells in the GI tract that exhibit the CD117+ immunophenotypes that is diagnostic of GIST.¹⁴ It is common to find these tumors anywhere along the gastrointestinal tract, including the stomach (40–60%), small intestine (30–40%), ano-rectum (7%), colon, and esophagus². However, greater omental GISTs have been reported in several studies by Zakaria et al,⁸ Franzini et al,⁹ Feng et al¹⁰ similar to our index case.

Generally, GIST is diagnosed at a median age of 60 years, and it's more common in men than in women, as indicated by Dimitri et al¹⁵ Franzini et al.⁹ However, our patient is 48 years old and female, a few years younger than 60 just as was also reported by Zakariya et al⁸. Also, according to a Medline search, 54 isolated cases or small series of omental GIST were reported^{3, 7, 16} among these cases, the median age at diagnosis is 65 years, male to female ratio is 1:1 and there was no difference in incidence between lesser and greater

omentum. These findings differed with study by Feng et al that reported 78/99 cases of omental GISTs that were seen in the greater omentum,¹⁴ and similar to our index case but differed from Dimitri et al that found theirs in lesser omentum.¹²

Despite their large size, omental GISTs can remain clinically silent. In most cases, patients present with an abdominal mass, but they are often diagnosed incidentally during investigations for other ailments. Similarly, our patient presented with progressive abdominal distension and weight loss that led to further evaluation, including differential diagnosis of GIST that was not considered originally. Examination, relevant investigations, and intraoperative findings revealed a greater omental mass to support abdominal distension, weight loss, constipation, but no history of vomiting was found. According to Sakurai et al, the normal omentum contains mesenchymal cells which are CD117+, similar to cajal cells, which are the source of the GIST⁹. We confirmed the diagnosis of our index case both histologically and by CD117+ immunohistochemistry. Our Patient was placed

on Imatinib 400mg daily based on immunohistochemistry positivity for CD117 among other post-operative medications. She

commenced regular follow up for 2 years and all her clinic visits have been uneventful.

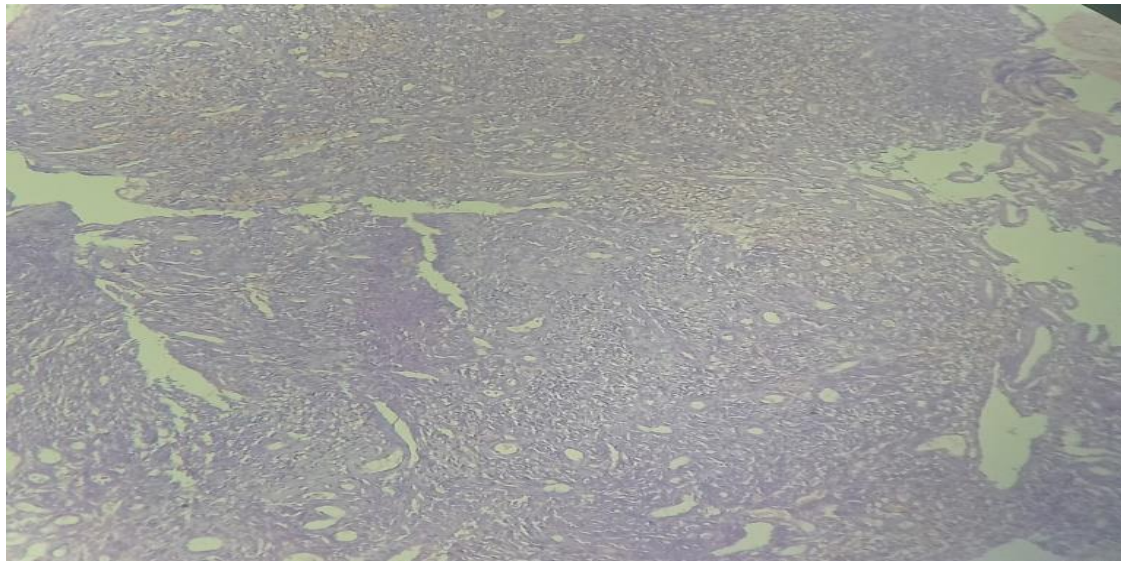


Figure 3: Histological section shows neoplastic cells that are made up of fascicles of spindle to ovoid cells with ovoid nuclei and mild atypical x 40.

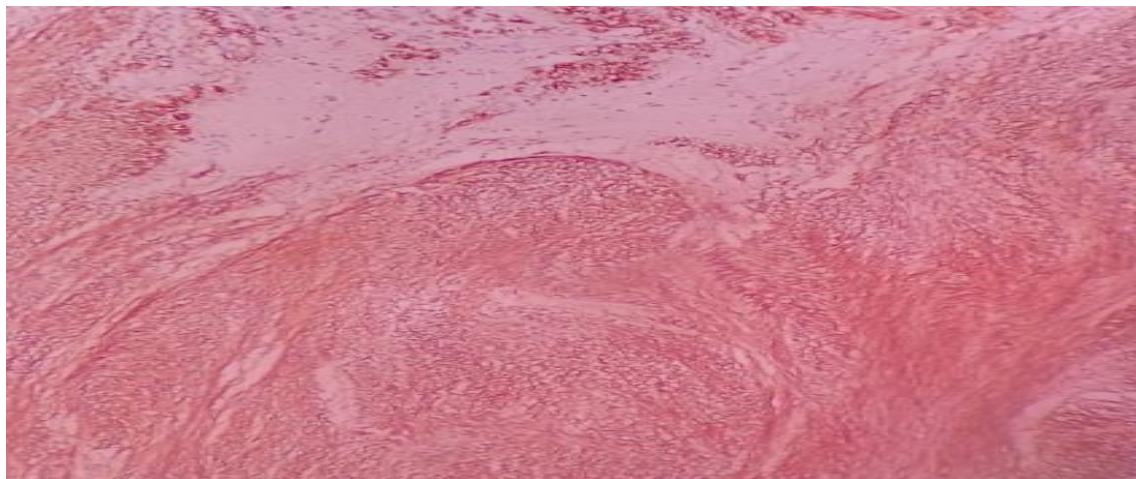


Figure 4: Shows GIST that are positive for CD117 x 40

CONCLUSION

There are very few GISTs that arise in the omentum, and surgical resection is the most effective treatment. In this index case, we used digital pathology tools in our primary and

secondary opinions to arrive at most diagnoses which enhance prompt and accurate diagnosis.

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Consent: oral consent was given.

Conflict of interest: none

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